

predictor of Claude usage (relevance) but being measured independently, is expected to be uncorrelated with sampling noise in our AUI estimates (validity). As noted above, states with more workers in high-Claude-usage roles do tend to have systematically higher usage per capita.

The 2SLS estimates imply modestly slower convergence: $\hat{\beta} \approx 0.89$ unweighted and $\hat{\beta} \approx 0.86$ when weighting by each state's working-age population. However, these estimates are less precise, and only the former is statistically distinguishable from 1 at the 10% level. Despite implying a slower convergence than OLS, the 2SLS estimates still imply rapid diffusion: just four to five years for the log deviation of each state's AUI to shrink by 90%.

That said, our estimates are based on just three months of data. And while the 2SLS specification may help address sampling noise, considerable uncertainty remains. We will revisit this question of the pace of diffusion in future reports.

¹ As with previous reports, all our analysis is based on privacy-preserving analysis. Throughout the report we analyze a random sample of 1M conversations from Claude.ai Free, Pro and Max conversations (we also refer to this as “consumer data” since it mostly represents consumer use) and 1M transcripts from our first-party (1P) API traffic (we also refer to this as “enterprise data” since it mostly represents enterprise use). Both samples come from November 13, 2025 to November 20, 2025. We continue to manage data according to our privacy and retention policies, and our analysis is consistent with our terms, policies, and contractual agreements. For 1P API data, each record is a prompt-response pair from our sample period which in some instances is mid-session for multi-turn interactions.

² The share of conversations on Claude.ai that were classified into neither automation nor augmentation categories fell from 3.9% to 3.0%.

³ See, for example, [Kalanyi et al \(2025\)](#): “Second, as the technologies mature and the number of related jobs grows, hiring spreads geographically. This process is very slow, taking around 50 years to disperse fully.”

⁴ With our bottom-up analysis of 1P API traffic we see Claude used to “Generate personalized B2B cold sales emails” (0.47%), “Analyze emails and draft replies for business correspondence” (0.28%), “Build and maintain invoice processing systems” (0.24%), “Classify and categorize emails into predefined labels” (0.23%), and “Manage calendar scheduling, meeting coordination, and appointment booking” (0.16%).

⁵ At a high level, we distinguish between automation and augmentation modes of using Claude. Automation encompasses interaction patterns focused on task completion: Directive: Users give Claude a task and it completes it with minimal back-and-forth; Feedback Loops: Users automate tasks and provide feedback to Claude as needed; Augmentation focuses on collaborative interaction patterns: Learning: Users ask Claude for information or explanations about various topics; Task Iteration: Users iterate on tasks collaboratively with Claude; Validation: Users ask Claude for feedback on their work.

⁶ These interaction modes are not mutually exhaustive. In some instances, Claude determines that a sampled conversation does not match any of the five interaction modes.

⁷ In this report we use Sonnet 4.5 for classification whereas in our previous Economic Index report we used Sonnet 4. We previously found that different models can generate different classification outcomes, though these effects tend to be modest.

⁸ We include a constant term in the regression since it should be equal to zero under the null hypothesis. Across all our specifications, the constant term is estimated to be close to and statistically indistinguishable from zero.

CHAPTER 2

Introducing economic primitives

The strength of the Anthropic Economic Index lies in showing not only how much AI is used, but how it is used. In prior reports, we showed which tasks Claude is used for, and how people collaborate with Claude. These data have enabled external researchers to analyze labor market shifts (e.g., [Brynjolfsson, Chandar & Chen, 2025](#)).

In this edition of the Anthropic Economic Index, we expand the breadth of data available to external researchers by providing insights on five economic “primitives”, by which we mean simple, foundational measures of the ways that Claude is used, which we generate by asking Claude to answer specific questions about the anonymized transcripts in our sample. Some of our primitives encompass several such questions, and others use a single indicator.

Because AI capabilities are advancing so rapidly and the economic effects will be unevenly experienced, we need a breadth of signals to uncover not just how Claude is used but also to inform what impact this technology will have.

Dimensions of AI use that matter for economic impacts

This report introduces five new economic primitives beyond the one we already measure, collaboration patterns (whether users automate or augment their tasks with Claude). These primitives capture five dimensions of a human-AI conversation: 1) task complexity, 2) human and AI skills, 3) work, coursework or personal use case, 4) the AI’s level of autonomy, and 5) task success (see Table 2.1). AI autonomy captures something different from our existing automation/augmentation distinction. For example, “Translate this paragraph into French” is high automation (directive, minimal back-and-forth) but low AI autonomy (the task requires little decision-making from Claude).

Economic primitives overview

Task complexity	Human time estimate	Estimate how many hours a competent professional would need to complete the tasks done by the Assistant. Assume they have: • The necessary domain knowledge and skills • All relevant context and background information • Access to required tools and resources • No access to AI tools to assist with the work
	Human with AI time estimate	Estimate how many minutes the User spent completing the tasks in the prompt with the Assistant. Consider: • Number and complexity of User messages • Time reading Assistant's responses • Time thinking and formulating questions • Time reviewing outputs and iterating • Realistic typing/reading speeds • Time implementing suggestions or running code outside of the conversation (only if directly relevant to the tasks)
	Multitasking	Did the User multitask in this conversation? Choose from these options: • Yes: the User was working on multiple tasks over the course of the conversation • No: the User was working on a single task over the course of the conversation
Human and AI skills	Human ability to complete task alone	Could the User have completed this task by themselves? Choose from these options: • Yes: the User would have been able to complete the task without the Assistant, even if it would have taken more time • No: the User would not have been able to complete the task without the Assistant, even with more time
	Human education years	Estimate how many years of formal education someone would need to understand the User prompts in this conversation. Your answer should be a single number out of the discrete numbers ranging from 0-20.
	AI education years	Estimate how many years of formal education someone would need to understand the Assistant responses in this conversation. Your answer should be a single number out of the discrete numbers ranging from 0-20.
Use case	Work vs. coursework vs. personal	Analyze whether the conversation between the User and the Assistant primarily focuses on work, coursework or personal use. Analyze the use case according to these categories: • Work: professional use to accomplish tasks that are part of the User's job • Coursework: use to help the User complete coursework in educational contexts • Personal: use for any domain that is not work or coursework
AI autonomy		Estimate how much autonomy the Assistant had to make decisions in this conversation (a discrete number ranging from 1-5, where 1 is none and 5 is extreme).
Task success		Did the Assistant complete the task provided by the User successfully? Choose from these options: • Yes: the Assistant completed the task provided by the User successfully • No: the Assistant did not complete the task provided by the User successfully

Table 2.1: Economic primitives added in this report.

The table shows the new economic primitives added in this report, beyond collaboration patterns (automation/augmentation) from prior reports. The first column shows the primitive category, the second column the name of the primitive, and the third column the operationalization of the primitives as the prompts provided to Claude which we use a classifier to map conversations to primitives. See online appendix at <https://huggingface.co/datasets/Anthropic/EconomicIndex> for full prompt texts.